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Div :- D Roll no. :- 26

course/Program :- S.Y BTech C.S.E

Subject :- DLM

Practical no. :- 03

* Aim :- Realization of half/full subtractor with logic gates.

* Theory :-

Subtractor are digital circuits that carry out subtraction of numbers. Subtractors are a key component of the ALU or the Arithmetic Logic unit. Binary subtraction is similar to that of decimal subtraction. Some basic binary subtractions are shown below:

$$\begin{array}{r} 0 \quad 0 \quad 1 \quad 1 \\ -0 \quad -1 \quad -0 \quad -1 \\ \hline 0 \quad 1 \quad 1 \quad 0 \end{array}$$

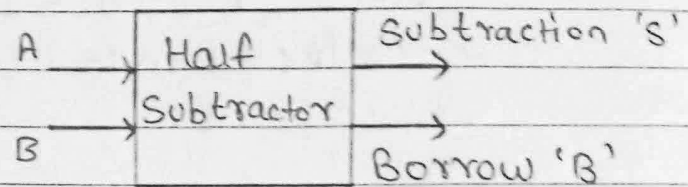
↳ Borrow

① Half Subtractor:

Half Subtractor is a combinational circuit that performs simple subtraction of two numbers specifically binary. If we assume A & B as the two bits where subtraction is to be done, the block diagram and Q truth table for half subtractor with A, B as inputs and subtraction and borrow as outputs can be tabulated as follows.



A	B	Subtraction	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



The Subtraction output of the binary Subtraction carried out above is similar to that of an Ex-OR.

The Borrow can be found out & Subtraction can be verified with the help of K-map.

Subtraction (S):

A \ B	0	1
0	0	1
1	1	0

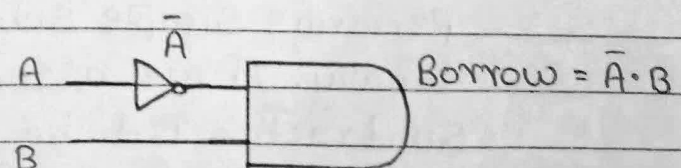
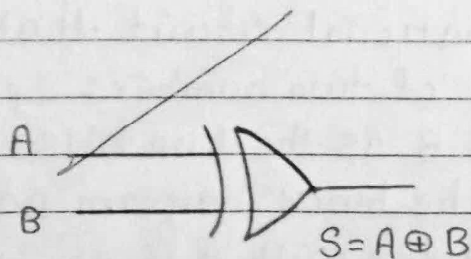
Borrow (B):

A \ B	0	1
0	0	1
1	0	0

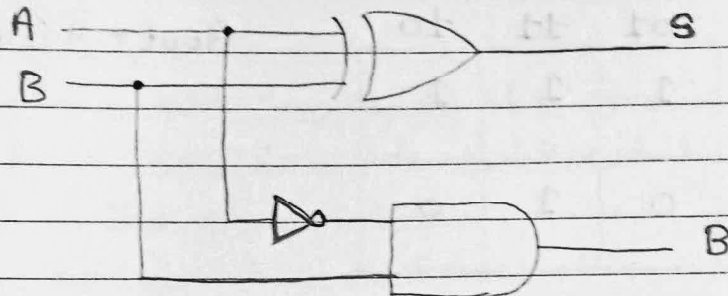
$$S = A\bar{B} + \bar{A}B$$

$$S = A \oplus B$$

$$\text{Borrow} = \bar{A} \cdot B$$



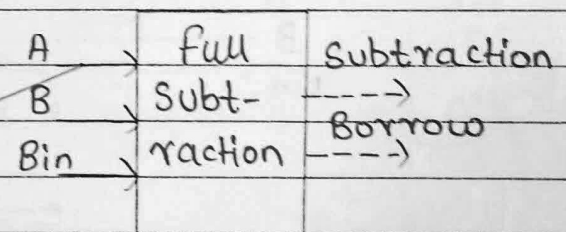
combining the logic circuits, we will get half subtractor.



② Full subtractor:

Full subtractor is a digital circuit used to calculate the subtraction of three binary bits. Full subtractors are complex and difficult to implement when compared to half subtractors. Two of the three bits are same as before which are A & B. The third bit is borrow bit from previous stage, represented as Bin. It calculates the subtraction of three bits along with the borrow.

A	B	Bin	Subtraction	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1



- Based on the truth table, the boolean function for Subtraction and borrow can be derived using K-map.



A \ B Bin	Borrow			
	00	01	11	10
0	0	1	1	1
1	0	0	1	0

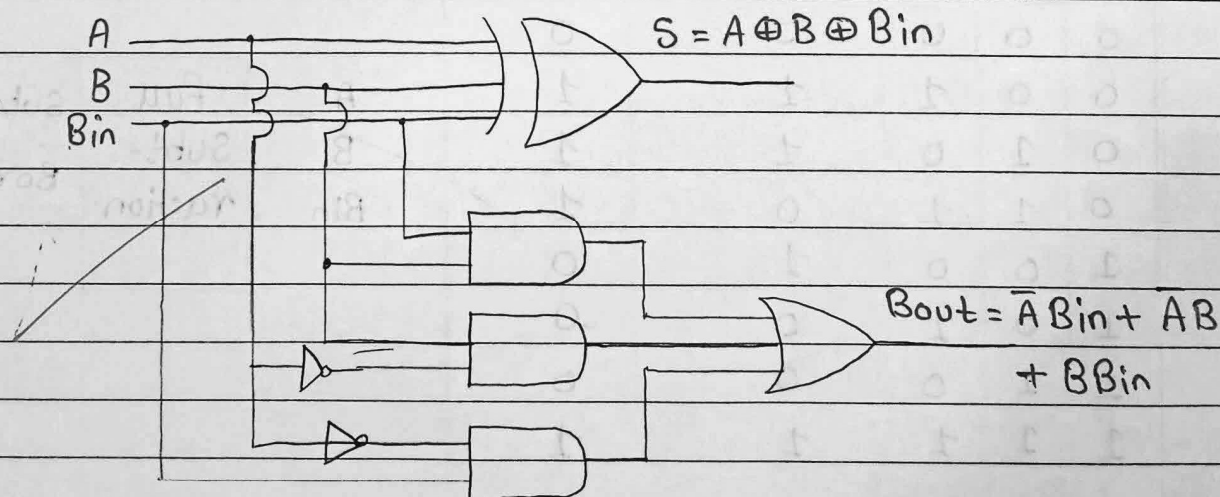
$$B_{out} = \bar{A} B_{in} + \bar{A} B + B B_{in}$$

A \ B Bin	Subtractor			
	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$S = \bar{A} \bar{B} B_{in} + A B B_{in} + A B B_{in} + A B B_{in}$$

$$\Rightarrow A \oplus B \oplus B_{in}$$

Logic diagram:





* Procedure:

1) Half Subtractor:

- Step: 1) Connect the +5V Supply to the circuit.
- Step: 2) First Press "SUBTRACT" to Subtract basic state of your output in the given table.
- Step: 3) Press the switches to select the required inputs A & B.
- Step: 4) Press the "SUBTRACT" button to subtract your inputs and outputs in the given table.

- Step: 5) Repeat Steps 3 & 4 for next state of inputs and their corresponding outputs.
- Step: 6) Press the "PRINT" button after completing your simulator to get your results.
- Step: 7) Press the "RESET" button whenever you want to replace your simulator.

2) Full Subtractor:

- Step: 1) Connect the +5V Supply to the circuit.
- Step: 2) First press "SUBTRACT" button to Subtract basic state of your output in the given table.
- Step: 3) Press the switches to select the required inputs 'A' & 'B' and 'Bin'.
- Step: 4) Press "SUBTRACT" button to subtract your Zlp and their output.
- Step: 5) Repeat Step: 3 & 4 for next state of Zlp and their corresponding outputs.
- Step: 6) Press the "PRINT" button after completing your Simulation to get results.



* Conclusion:

The experiment successfully implemented half and fully subtractor circuits using logic gates. Both circuits accurately computed the difference and borrow, matching the theoretical truth tables and confirming their correct operation.

~~Kulkarni~~